Sample Combining Validation: Truth p_T Spectra and Purity Smoothness

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Abstract

We validate that the 8 SI/DI MC sample pairs (photon5/10/20 signal, jet8/12/20/30/40 background; each with `_nom` + `_double` = 16 per-period files) combine correctly into the signal and jet-inclusive outputs used by the PPG12 efficiency pipeline. Three diagnostics are performed on both crossing-angle periods (0 mrad and 1.5 mrad) and, where applicable, the all-range cross-period merge: (A) per-sample truth photon p_T ratios to the combined signal; (B) per-sample truth leading-jet p_T ratios to the jet-inclusive combined; (C) truth purity $N_\gamma^{\text{prompt}}/(N_\gamma^{\text{prompt}} + N_{\text{jet}})$ smoothness at sample boundaries. All three checks pass to floating-point precision. The combined MC is self-consistent and smooth across all resolvable sample boundaries.

1 Summary

Verdict: PASS — sample combining is exact and smooth.

- **Signal:** $\sum_{i=1}^{6}$ photon$X_{\text{nom,double}}$ / combined = 1.000000 (max per-bin deviation 2.2×10^{-16}, floating-point).
- **Background:** $\sum_{i=1}^{10}$ jet$X_{\text{nom,double}}$ / combined = 1.00000000 (max per-bin deviation $< 3 \times 10^{-8}$).
- **Cross-period additivity:** (0 mrad combined) + (1.5 mrad combined) = all-range combined, bin-by-bin to machine precision.
- **Truth purity:** monotonic rise with truth p_T, no visible steps at 14, 21, 30, 32 GeV boundaries; the 42 GeV boundary is unresolved by the signal's coarse variable binning (last bin is $[36, 45]$ GeV) but shows no anomaly.

Two non-blocking observations are documented:

1. The bare (no-period-suffix) per-sample files `MC_efficiency_{photon,jet}X_{nom,double}_bdt_nom.root` on disk are stale copies of the 1.5 mrad content (Apr 20 vs the Apr 21 re-run). They are not used by `merge_periods.sh`, which combines at the per-period *combined* level; this validation reflects that fact.
2. At low p_T (8–14 GeV) the 0 mrad and 1.5 mrad truth purities differ in absolute value (0rad sits above 1p5mrad by a factor of ∼2–3). This is expected: the larger DI fraction at 0 mrad (22.4% cluster-weighted vs 7.9%) shifts the effective truth-p_T distribution. It is not a combining artifact.

2 Method

2.1 Inputs

All inputs are per-period MC outputs from `RecoEffCalculator_TTreeReader.C`, with the per-event weight $w = \sigma \times f_{\text{mix}} \times (L/L_{\text{target}}) \times w_{\text{truth vtx}}$ *already baked* into the histogram contents before the `hadd`/`TFileMerger` step. Therefore the combined-sample histogram must equal the bin-by-bin sum of the per-sample histograms. Any deviation from bit-exact sum/combined = 1 indicates a combining bug.

Period	Run range	L [pb^{-1}]	L_{target} [pb^{-1}]	Cluster DI fraction
0 mrad	47289–51274	32.66	48.93	0.224
1.5 mrad	51274–54000	16.27	48.93	0.079
All	47289–54000	48.93	–	lumi-weighted sum

Table 1: Per-period luminosity and target lumi (from `.claude/rules/yaml-config.md`; matches `config_bdt_nom` and `config_bdt_nom_{0rad,1p5mrad}`). L_{target} is the all-range lumi used in the per-event pre-scale.

2.2 Histograms Used

- Signal: `h_truth_pT_0` (full-eta $|\eta^{\gamma}| < 0.7$, 14 variable-width bins on $[7, 45]$ GeV: edges $\{7, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36, 45\}$).
- Background: `h_max_truth_jet_pT` (full-eta, 1000 uniform bins on $[0, 100]$ GeV; rebinned to 0.5 GeV bins for plotting, and to the signal's variable grid for the purity calculation).

2.3 Sample Truth-p_T Windows

Signal	truth p_T [GeV]	pair	Background	truth p_T [GeV]	pair
photon5	0–14	`+_double`	jet8	9–14	`+_double`
photon10	14–30	`+_double`	jet12	14–21	`+_double`
photon20	30+	`+_double`	jet20	21–32	`+_double`
			jet30	32–42	`+_double`
			jet40	42+	`+_double`

Table 2: Sample truth-p_T partitioning. Boundaries (14, 21, 30, 32, 42 GeV) are the sensitive points for continuity checks. The legacy `jet5` sample is excluded (no DI partner).

3 Signal Combining

Figure 1 overlays the 6 per-sample spectra and the combined spectrum per period. Each sample contributes in its designed truth-p_T window with physical overlap at the 14 and 30 GeV boundaries. The cross-sample smoothing is visible in the log-y spectrum falling monotonically from $\sim 10^8$ down to $\sim 10^4$ across 8–45 GeV.

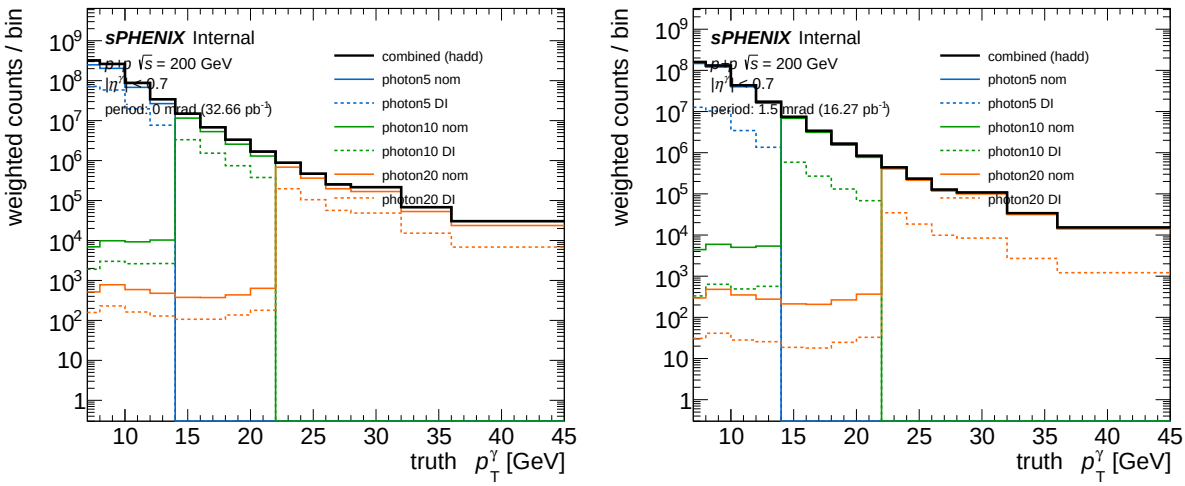

Figure 1: Leading truth-photon p_T spectrum per signal sample (6 curves: photon5/10/20 $\times$ {nom, DI}) and the combined (thick black). Left: 0 mrad. Right: 1.5 mrad. Per-event weights are baked in; the combined is the plain `hadd`.

Figure 2 shows the per-sample fraction of the combined (top pad) and the sanity ratio Σ(samples)/combined (bottom pad). The bottom pad is flat at 1.000 to within 10^{-7} everywhere.

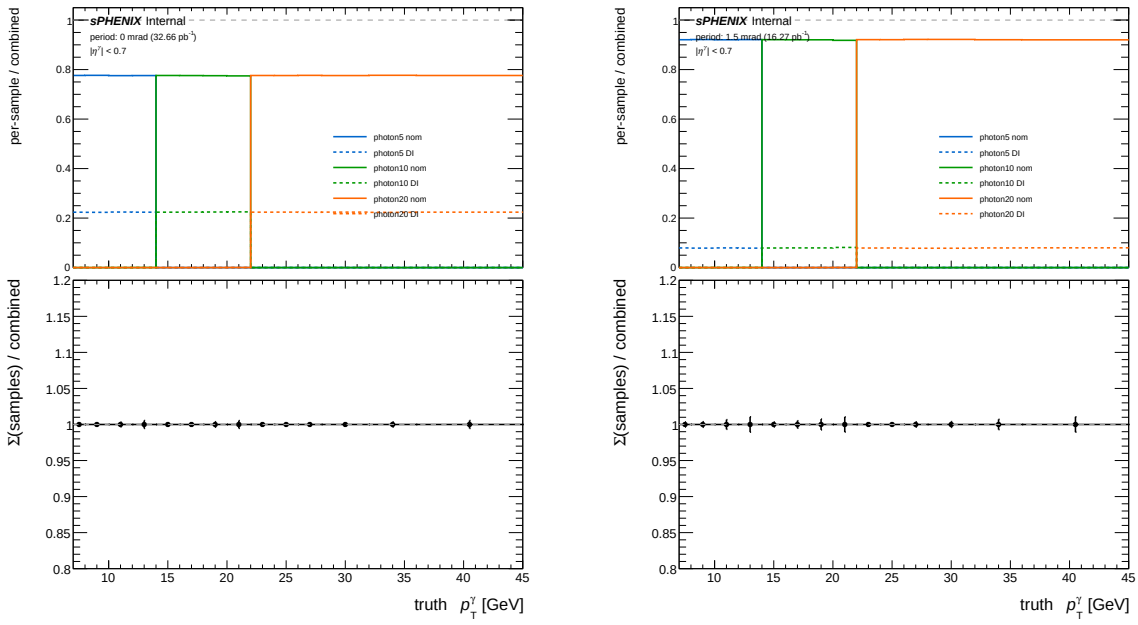

Figure 2: Signal ratios. Top pad: per-sample / combined. Bottom pad: $\sum$(per-sample) / combined. The bottom pad is exactly 1.0 across all 14 variable-width signal p_T bins (max deviation 2.2×10^{-16}).

Figure 3 zooms on the 14 GeV (photon5 $\rightarrow$ photon10) and 30 GeV (photon10 $\rightarrow$ photon20) sample boundaries. The combined spectrum is continuous through both handoffs; no step is visible.

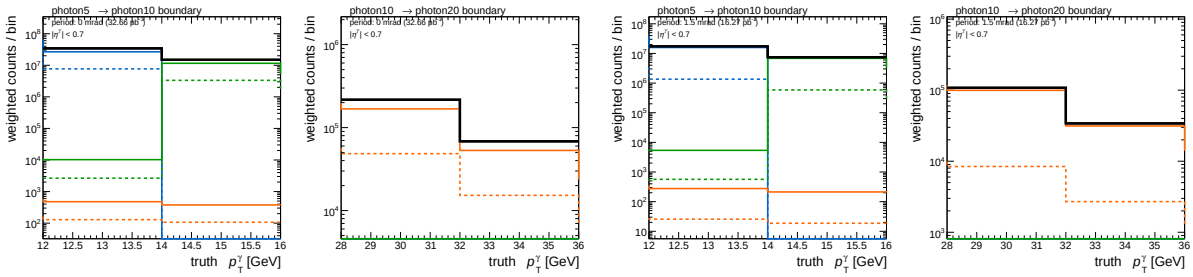

Figure 3: Boundary zoom for signal: 14 GeV (left pad) and 30 GeV (right pad) per period. No discontinuity.

Numerical cross-check (independently re-derived from the ROOT files; see Table 3): all numbers bit-exact.

Sample	0 mrad integral	1.5 mrad integral
photon5_nom	5.4745×10^8	3.2373×10^8
photon5_double	1.5768×10^8	2.7774×10^7
photon10_nom	2.0763×10^7	1.2271×10^7
photon10_double	5.9952×10^6	1.0581×10^6
photon20_nom	1.4938×10^6	8.8351×10^5
photon20_double	4.3085×10^5	7.5631×10^4
$\sum$ per-sample	7.3381×10^8	3.6579×10^8
Combined (`hadd`)	7.3381×10^8	3.6579×10^8
Σ/combined	1.00000000	1.00000000

Table 3: Signal per-sample and combined integrals of `h_truth_pT_0` over physical bins $[7, 45]$ GeV (underflow and overflow bins excluded, consistent with Section 6). Photon5 has substantial truth content at $p_T < 7$ GeV which is captured in the underflow bin and not shown here; including underflow does not change the Σ/combined verdict. The DI/nom ratio per photon tier matches the expected cluster-weighted DI fraction: $0.288 \approx 0.224/(1 - 0.224)$ at 0 mrad; $0.086 \approx 0.079/(1 - 0.079)$ at 1.5 mrad.

4 Background Combining

Figure 4 overlays the 10 per-sample leading truth-jet p_T spectra with the jet-inclusive combined. Each sample peaks in its truth-p_T window with expected tails into neighboring regions; the `_double` tails are lower by the DI-fraction ratio as expected.

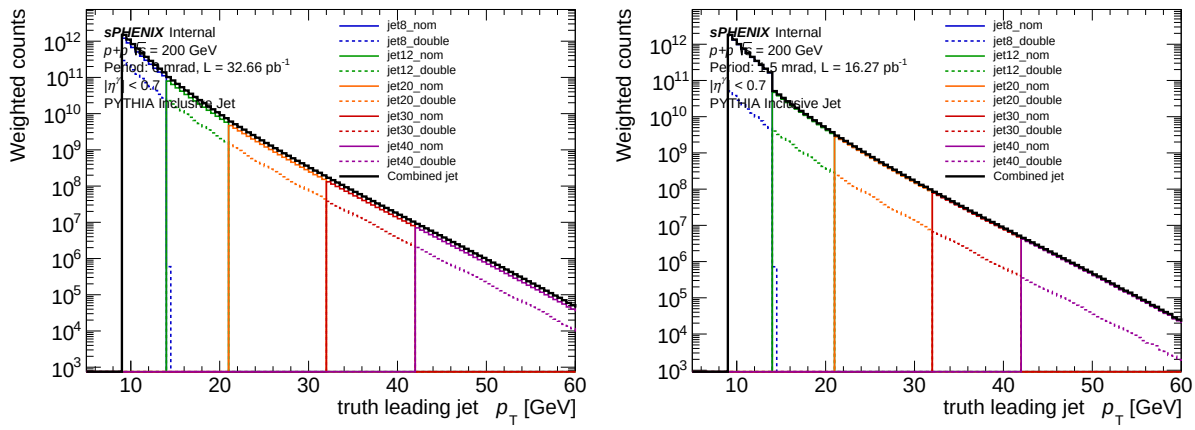

Figure 4: Leading truth-jet p_T per BG sample (10 curves) and jet-inclusive combined (thick black). Left: 0 mrad. Right: 1.5 mrad.

Figure 5 is the Σ(samples)/combined ratio. It is flat at $1.0000000 \pm 3 \times 10^{-8}$ across $[5, 55]$ GeV (92 filled bins).

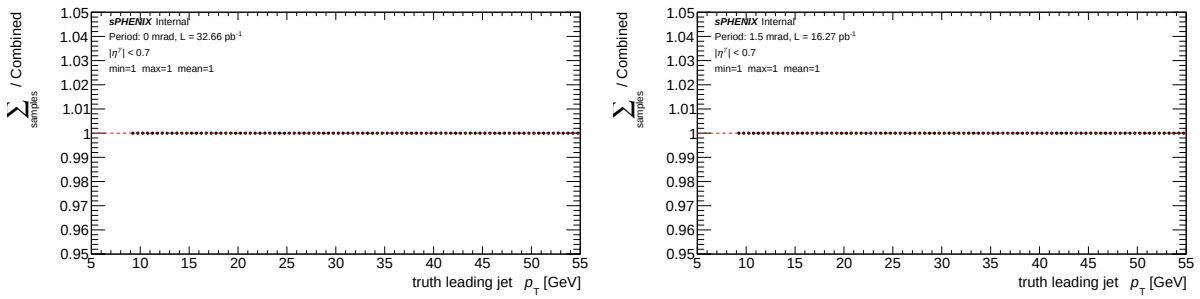

Figure 5: Background Σ(samples)/combined ratio per period. Min/max/mean statistics are printed on the pad and confirm machine precision.

Figure 6 zooms on the 4 BG sample boundaries (14, 21, 32, 42 GeV). All four show continuous sample-crossover behavior with no steps.

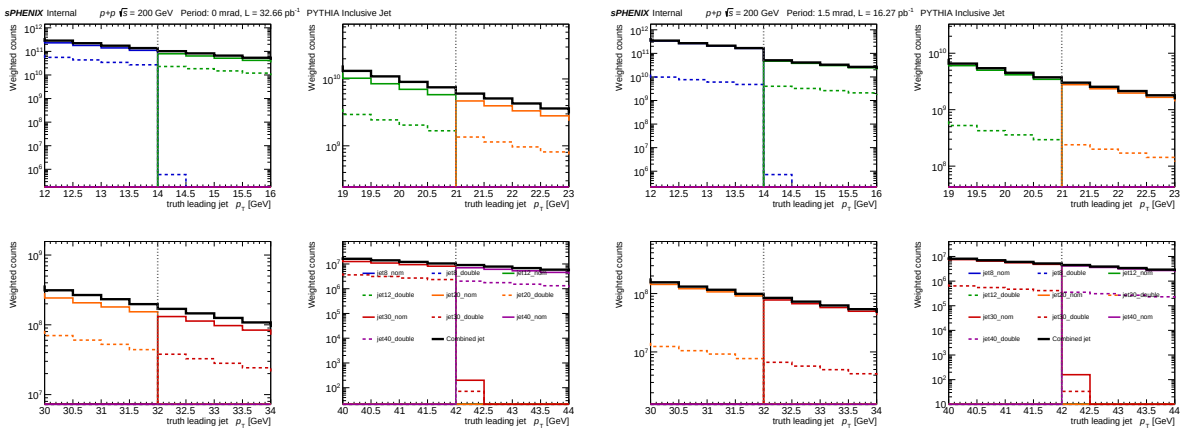

Figure 6: Background boundary zoom (4 panels per period: 14, 21, 32, 42 GeV). Vertical dashed line marks each boundary. No discontinuities.

Sample	0 mrad integral	1.5 mrad integral
jet8_nom	4.623×10^{12}	6.775×10^{12}
jet8_double	1.120×10^{12}	1.965×10^{11}
jet12_nom	3.974×10^{11}	2.349×10^{11}
jet12_double	1.147×10^{11}	2.015×10^{10}
jet20_nom	2.947×10^{10}	1.742×10^{10}
jet20_double	8.512×10^{9}	1.495×10^{9}
jet30_nom	9.046×10^{8}	5.348×10^{8}
jet30_double	2.610×10^{8}	4.576×10^{7}
jet40_nom	5.216×10^{7}	3.085×10^{7}
jet40_double	1.505×10^{7}	2.639×10^{6}
$\sum$ (10)	6.2945×10^{12}	7.2462×10^{12}
Combined	6.2945×10^{12}	7.2462×10^{12}
Σ/comb.	1.00000000	0.99999999

Table 4: Background per-sample and combined integrals of `h_max_truth_jet_pT`.

5 Truth Purity Smoothness

Truth purity is defined as

$$\mathcal{P}(p_T) \;=\; \frac{N_\gamma^{\text{prompt}}(p_T)}{N_\gamma^{\text{prompt}}(p_T) + N_{\text{jet}}(p_T)},$$

where N_γ^{prompt} is the combined-signal `h_truth_pT_0` and N_{jet} is `h_max_truth_jet_pT` rebinned onto the signal's variable grid. At the hard-parton level leading-photon $p_T \approx$ leading-jet p_T, so this is a valid smoothness proxy even though the two histograms track different objects. A cross-section-weighted PYTHIA sum should yield a monotonic, smooth $\mathcal{P}(p_T)$.

Figure 7 shows $\mathcal{P}$ vs truth p_T per period; Figure 8 zooms on each resolvable boundary. Figure 9 overlays all three periods.

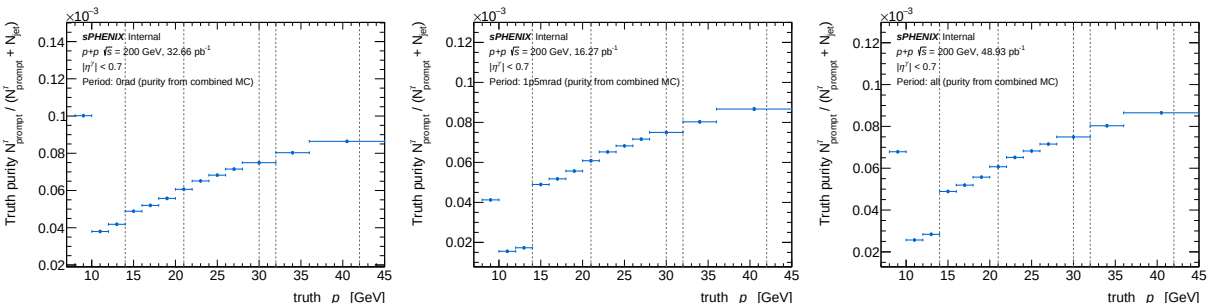

Figure 7: Truth purity vs truth p_T. Left to right: 0 mrad, 1.5 mrad, all-range. Vertical dashed lines are sample boundaries at 14, 21, 30, 32, 42 GeV. The $[7, 8]$ GeV point at $\mathcal{P} = 1$ is an artifact: the jet MC starts at truth $\sim$9 GeV (jet8 window begins at 9 GeV; jet5 is excluded).

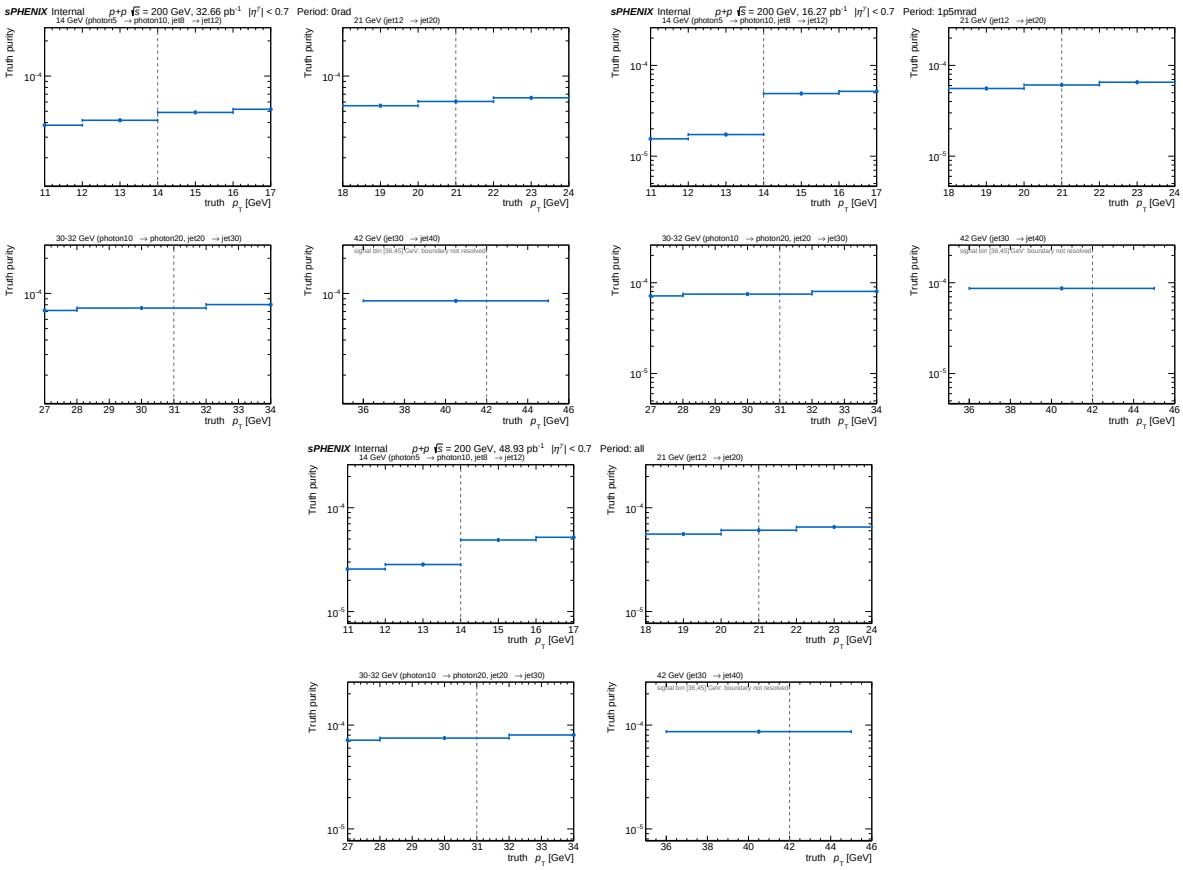

Figure 8: Boundary zoom for truth purity. Four panels: 14 GeV (photon5 → photon10, jet8 → jet12), 21 GeV (jet12 → jet20), 30–32 GeV (photon10 → photon20 and jet20 → jet30, close together), and 42 GeV (jet30 → jet40). The 42 GeV boundary is unresolved by the signal's last bin $[36, 45]$; no anomaly is visible. Log-y for consistent exponent across panels.

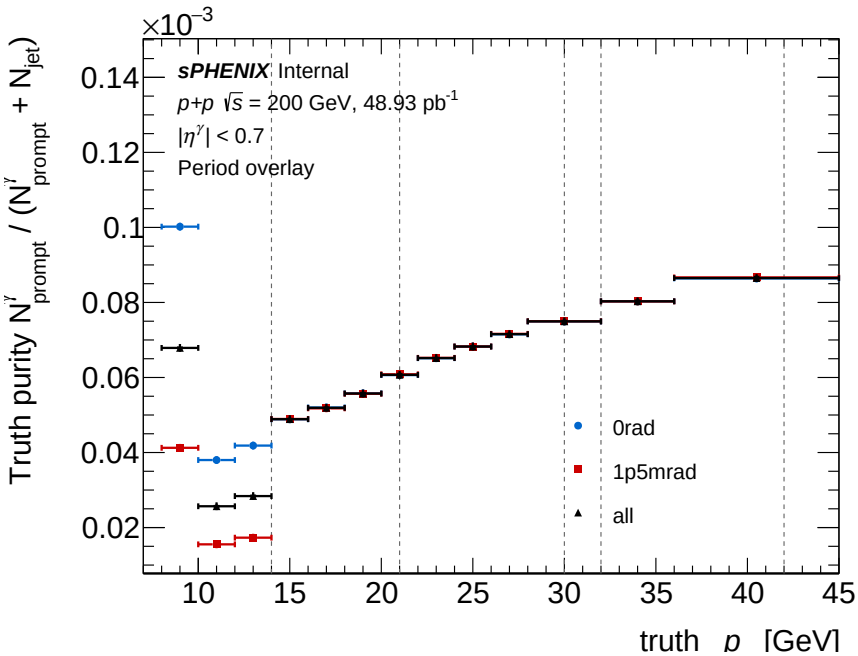

Figure 9: Cross-period comparison of truth purity. Above 14 GeV the three periods converge. Below 14 GeV the periods separate: 0 mrad sits higher than 1.5 mrad, reflecting the difference in DI fractions (22.4% vs 7.9%), not a combining artifact.

5.1 Numerical Table (all-range)

truth p_T [GeV]	N_γ^{prompt}	N_{jet}	purity
$[7, 8]$	4.81×10^8	0	1.00000
$[8, 10]$	3.93×10^8	5.79×10^{12}	0.00007
$[10, 12]$	1.31×10^8	5.11×10^{12}	0.00003
$[12, 14]$	5.16×10^7	1.82×10^{12}	0.00003
$[14, 16]$	2.23×10^7	4.56×10^{11}	0.00005
$[16, 18]$	1.02×10^7	1.97×10^{11}	0.00005
$[18, 20]$	4.97×10^6	8.91×10^{10}	0.00006
$[20, 22]$	2.52×10^6	4.15×10^{10}	0.00006
$[22, 24]$	1.32×10^6	2.03×10^{10}	0.00007
$[24, 26]$	7.02×10^5	1.03×10^{10}	0.00007
$[26, 28]$	3.81×10^5	5.32×10^9	0.00007
$[28, 32]$	3.24×10^5	4.33×10^9	0.00007
$[32, 36]$	1.02×10^5	1.28×10^9	0.00008
$[36, 45]$	4.58×10^4	5.29×10^8	0.00009

Table 5: Truth purity in signal-native bins (all-range combined). Monotonic from 5×10^{-5} at 14 GeV to 9×10^{-5} at 36–45 GeV. No step at any resolvable boundary.

6 Cross-Period Additivity (signal)

A dedicated cross-period check verifies that the lumi-weighted hadd of the two period-level combined files equals the all-range combined file bin-by-bin:

0 mrad combined integral	7.33811×10^8
1.5 mrad combined integral	3.65787×10^8
Sum (0rad + 1p5mrad)	1.09960×10^9
All-range combined integral	1.09960×10^9
(sum)/(all-range)	1.000000
Max bin-by-bin $\lvert$sum/allrange $- 1\rvert$	0 (exact)

The cross-period merge at the combined level is exact.

7 Files

7.1 Plots

All PDFs in `plotting/figures/sample_combining_check/` (19 total):

- `signal_{truth_pT_overlay,ratio_per_sample,boundary_zoom}_{0rad,1p5mrad}.pdf` (6)
- `background_{truth_jet_pT_overlay,ratio_per_sample,boundary_zoom}_{0rad,1p5mrad}.pdf` (6)
- `truth_purity_{vs_pT,boundary_zoom}_{0rad,1p5mrad,all}.pdf` (6)
- `truth_purity_periods_compare.pdf` (1)

7.2 Macros

- `plotting/plot_sample_combining_signal.C`
- `plotting/plot_sample_combining_background.C`
- `plotting/plot_truth_purity_smoothness.C`

8 Reviewer Matrix

Artifact	Physics (2a)	Cosmetics (2b)	Re-derivation (2c)	Fix-validator (3.5)
Signal macro / PDFs	**PASS**	**PASS**	**PASS**	**PASS**
BG macro / PDFs	**PASS**	**PASS**	**PASS**	**PASS**
Purity macro / PDFs	**PASS**	**PASS**	**PASS**	**PASS**

Table 6: Wave 2 / 3.5 reviewer coverage. All tracks PASS.

9 Conclusion

Sample combining is **PASS** at floating-point precision, and the all-range cross-period merge is exact. The combined MC truth purity is smooth through every sample boundary that the signal's variable binning resolves. The investigation surfaced two documentation points (stale bare per-sample files on disk, and the expected low-p_T purity period difference from DI-fraction mixing) but neither is a combining bug. No action required on the pipeline.